

SOWMIYALAKSHMI GANESH

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EDUCATION

University of California Irvine – Master of Data Science

Sep 2024 – Dec 2025

Generative AI, Machine Learning, Deep Learning, Data Mining, Statistics, Data Structures

CGPA: 3.93/4.0

Vellore Institute of Technology – Bachelor of Technology in Computer Science

Sep 2020 – May 2024

Database Management, Cloud Computing, Data Visualization, Web Development

CGPA: 3.89/4.0

TECHNICAL SKILLS

Languages & Frameworks: Python, SQL, R, REST API, React.js, Node.js, Express.js

Databases & Storage: DynamoDB, PostgreSQL, MySQL, MongoDB, Amazon S3

Cloud, DevOps & Deployment: AWS (Fargate, CodePipeline), Microsoft Azure, Docker, CI/CD, Git, GCP

AI/ML: Agentic AI (LangGraph, MCP), RAG, LLM ops (vLLM, QLoRA), PyTorch, TensorFlow, scikit-learn, Pandas, MLflow

Certifications: AWS Certified Solutions Architect – Associate

EXPERIENCE

Zof AI

Jun 2025 – Present

Software Developer Intern - AWS, Agentic AI, React.js, LangGraph, PostgreSQL, Docker

San Francisco, California

- Designed an Agentic AI pipeline with LangGraph and FastAPI microservice backend enabling automated API testing that reduced overall testing effort by 90%.
- Refactored the legacy front-end using React to obtain fine-grain control of UI/UX components and responsive design.
- Implemented thread-safe circular buffer to enable efficient, real-time and scalable log aggregation.
- Integrated JSON output logs from the agents having different formats with an MCP server, generating standardized formats for downstream database storage.
- Analyzed backend test execution database using advanced SQL operations to generate insights on system performance trends.
- Containerized and deployed the pipeline on AWS Fargate using Docker with CI/CD automation through CodePipeline and ECR and enabled log storage in AWS S3.

Calit2 - UC Irvine Division

Mar 2025 – May 2025

AI Developer Intern - Python, Pytorch, MLflow, RAG, QLoRA, Cuda

Irvine, California

- Fine-tuned BERT-large-uncased using QLoRA on Bloom's Taxonomy classification with PyTorch and Cuda acceleration, achieving 71.9% test accuracy with MLflow experiment tracking for reproducible hyperparameter optimization.
- Benchmarked against RAG-based approach using vLLM-accelerated Mistral-7B for semantic query compression and Pinecone vector database optimization, demonstrating 68.4% accuracy with 40% embedding storage reduction.

VIT Transit

May 2023 – Jul 2023

Software Developer Intern - Python, Tensorflow, OpenCV, Scikit-image

Vellore, India

- Developed a responsive shuttle tracking web application using React that improved the tracking efficiency by 25%.
- Built backend services using Node.js, Express.js, and MongoDB/MySQL for user authentication, booking & scheduling routes.
- Integrated UPI payment gateway for shuttle booking payments and implemented Google Maps API for live location tracking.

PROJECTS

EzPlanner - React, Node.js, JavaScript, MongoDB, GraphQL, Express.js, JWT

Jan 2023 – Mar 2023

- Developed a full-stack event management platform using React.js and Apollo GraphQL, implementing type-safe queries for efficient data fetching and real-time event updates.
- Built scalable backend API using Apollo Server and Express.js with MongoDB database, designed GraphQL resolvers for user authentication, event CRUD operations and booking management.
- Created responsive front-end interface using React.js, Material-UI components, and Apollo Client for optimized caching and state management.
- Implemented secure user authentication system with JWT tokens and role-based access control, integrated Mongoose ODM for data validation.

QueryGen AI: Natural Language to SQL Generator 📄 [Link](#)

Nov 2023 – Dec 2023

- Contributed to an intuitive RAG based full-stack application for automated database query generation and execution.
- Created vector embeddings of tables present in database schema using Langchain, stored the results in Pinecone database.
- Performed similarity search of the user's query with the generated vector embeddings to find the top 5 relevant tables.
- Integrated OpenAI's language model to generate SQL/NoSQL queries by providing the model with the user's prompt along with the relevant tables.